Chaitanya Jindal

Machine Learning Engineer

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I build production-focused AI systems across computer vision, retrieval, and data pipelines. At Droisys, I developed computer vision tooling that reduced manual labeling effort by 60% and dataset preparation time by 40%. My work spans data preparation, model development, retrieval systems, and production tooling.

EDUCATION

Post Graduate Certificate Programme in Artificial Intelligence **February 2026 - August 2026**

Centre for Development of Advanced Computing, Noida

- CCAT Rank: AIR 286
- Intensive 6-month programme covering Deep learning, Computer Vision, NLP, and MLOps, with hands-on labs in PyTorch, TensorFlow and cloud deployment.

Bachelor of Technology (B.Tech) Computer Science and Engineering: 78.1% **November 2025**

JSS Academy of Technical Education, Gautam Buddh Nagar

SKILLS

Languages:	Python, SQL, C++, JavaScript
AI/ML:	PyTorch, TensorFlow, YOLO, OpenCV, LangChain
Retrieval & GenAI:	GraphRAG, Neo4j (Cypher), Qdrant, BGE-M3, GLiNER, Cross-Encoder Reranking
Data & MLOps:	Docker, MLflow, PySpark, Git
Frameworks:	FastAPI, Django, PyQt

EXPERIENCE

Associate Data Analyst **April 2025 - February 2026**

Droisys *Noida, India*

- Developed a custom "Auto-Annotator" toolkit using Python and localized template matching, reducing manual labeling labor by 60%.
- Engineered an automated image cropping pipeline utilizing YOLO11 and OpenCV, reducing dataset preparation time by 40% and accelerating the ML production lifecycle.
- Designed and deployed automated data validation scripts using Pandas to sanitize large-scale preproduction datasets, ensuring 100% data quality for downstream Computer Vision models.

PROJECTS

GraphReg: Hybrid GraphRAG Platform for Regulatory Compliance

- Co-developed a hybrid Graph + Vector RAG platform for SEBI regulatory text; owned the backend graph and retrieval layer, combining Neo4j 2-hop graph traversal with top-k semantic search over Qdrant using BGE-M3 embeddings.
- Collaborated on layout-aware parsing (IBM Docling) and zero-shot entity extraction (GLiNER), then added cross-encoder reranking (ms-marco-MiniLM) to retrieval pipeline, powering citation-backed, auditable outputs in the React UI

QuickDraw: Smart Reference Annotator

- Developed an end-to-end desktop annotation tool using PyQt6, integrating SIFT/FLANN feature matching for semi-automatic bounding box generation.
- Engineered a persistent HDF5 database to store feature descriptors, allowing the Auto-Annotate feature to become functional with only 10 manual examples, thus reducing human review time by over 70%.

BEYOND WORK

- Built and self-hosted a home lab with a Raspberry Pi as the central node, that handles NAS, real-time data ingestion, and media streaming across all devices. Basically my own private cloud.